XAI: Model-Agnostic Methods

Exercise 5 — Partial Dependence Plots (PDP)

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1. Introduction

This report presents a comprehensive analysis of Partial Dependence Plots (PDPs), a widely adopted model-agnostic interpretability technique that allows practitioners to understand how individual features influence model predictions.

2. Methodology: Partial Dependence Plots

A Partial Dependence Plot (PDP) visualizes the marginal effect of one or two features on the predicted outcome of a previously fitted model. Formally, for a feature of interest xS, the partial dependence function is defined as:

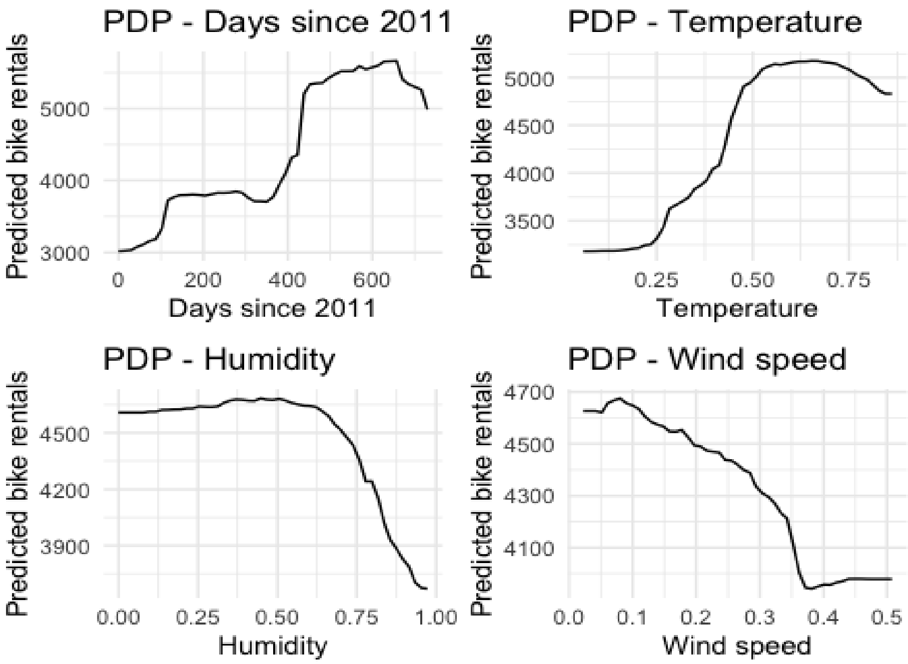
$$\hat{f}_{x_S}(x_S) = E_{x_C}\left[\hat{f}(x_S, x_C)\right] = \int \hat{f}(x_S, x_C) d\mathbb{P}(x_C)$$

PDPs are classified as global interpretability methods, as they describe the overall relationship between a feature and the model output across the entire dataset, rather than focusing on individual predictions. They can reveal whether a relationship is linear, monotonic, or more complex, and can expose non-linearities that simpler models would fail to capture.

An important assumption underlying PDPs is that the feature of interest is not strongly correlated with the remaining features. When this assumption is violated, the marginal averaging procedure may evaluate the model at unrealistic combinations of feature values, potentially leading to misleading interpretations. In the present analysis, this limitation is acknowledged, and results are interpreted with appropriate caution.

For the two-dimensional PDP in Section 4, both temperature and humidity are studied simultaneously, producing a heatmap that captures their joint effect on the predicted outcome. This extension allows the detection of interaction effects that would be invisible in separate one-dimensional plots.

3. Exercise 1 — One-Dimensional PDP: Bike Rental Demand



The Random Forest was trained with 500 trees and a fixed random seed (42) to ensure reproducibility. The model captures complex non-linear interactions between weather conditions and rental demand relationships that a linear model would be unable to represent adequately. Partial Dependence Plots were then computed for four key variables using the pdp library, with predictions averaged across the full training dataset.

3.1 Results and Interpretation

Days Since 2011

The PDP for days since 2011 reveals a clear and sustained upward trend in predicted bike rentals over the two-year observation window. At the beginning of the period, the model predicts approximately 3,000 daily rentals, rising to over 5,000 rentals by mid-2012. This growth trajectory reflects the increasing adoption of the bike-sharing service over time, likely driven by a combination of factors: greater public awareness of the system, infrastructure expansion, promotional campaigns, and the normalization of cycling as an urban commuting option. Notably, a slight decline appears towards the very end of the period, which may be attributable to seasonal effects in late 2012 that the temporal variable partially absorbs. Overall, this variable captures a structural growth trend in demand rather than a cyclical or weather-driven pattern.

Temperature

Temperature emerges as the most influential meteorological predictor in the model. The PDP shows a strongly positive relationship between normalized temperature and predicted rentals across most of the temperature range, with predictions rising from approximately 3,200 rentals at low temperatures to a peak of around 5,100 rentals at medium-high temperatures (normalized value ~0.55). Beyond this peak, the curve plateaus and declines slightly, suggesting that extreme heat reduces the attractiveness of cycling — a finding consistent with behavioral intuition, as very hot days can make outdoor physical activity uncomfortable or even hazardous. This non-linear, inverted-U pattern could not be captured by a linear model, illustrating the value of both the Random Forest and the PDP as analytical tools.

Humidity

The PDP for humidity demonstrates a largely flat relationship at low-to-moderate humidity levels (normalized values 0.0 to approximately 0.65), where predicted rentals remain stable around 4,600–4,700 per day. However, above a humidity threshold of roughly 0.70, the model predicts a sharp and sustained decline in demand, dropping to approximately 3,800 rentals at maximum humidity. This threshold effect suggests that users tolerate moderate humidity without significantly altering their behavior, but respond strongly to high humidity — likely associated with rainy, damp, or muggy conditions that make cycling unpleasant. From a business perspective, this insight could inform dynamic pricing or availability adjustments on high-humidity days.

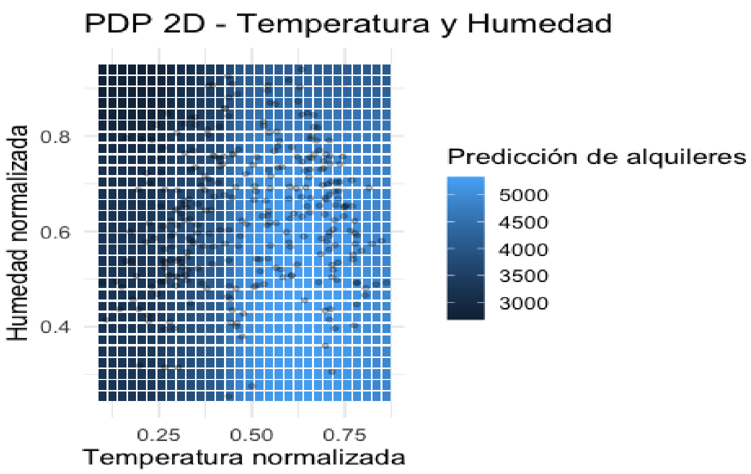
Wind Speed

Wind speed displays a monotonically negative relationship with bike rental demand. The PDP shows that even at low wind speeds, predicted rentals are already declining from their peak (~4,700), and this decline accelerates at higher wind speeds (normalized values above 0.30), where rentals fall to approximately 4,050. The effect is consistent and interpretable: strong winds increase the physical effort required to cycle and can pose safety concerns, particularly for casual or recreational riders. Unlike humidity, which only exerts a strong negative effect above a threshold, wind speed appears to continuously erode demand throughout its range.

3.2 Summary

Across the four variables analyzed, temperature exerts the strongest and most nuanced influence on bike rental demand, exhibiting a non-linear relationship with an optimal range around moderate warmth. Days since 2011 captures a positive growth trend in system adoption. Humidity and wind speed both reduce demand, with humidity acting primarily as a threshold effect and wind speed exerting a more continuous negative influence. These findings are consistent with physical and behavioral intuition and provide actionable intelligence for operators of the bike-sharing system.

4. Exercise 2 — Two-Dimensional PDP: Temperature and Humidity



Given the computational cost of evaluating the Random Forest at every combination of values in a two-dimensional grid, a random subsample of 300 observations was drawn from the dataset before computing the PDP. This subsampling step is consistent with best practices for 2D PDPs and does not substantially alter the interpretation, as the dataset is relatively homogeneous in its distributional properties.

4.2 Results and Interpretation

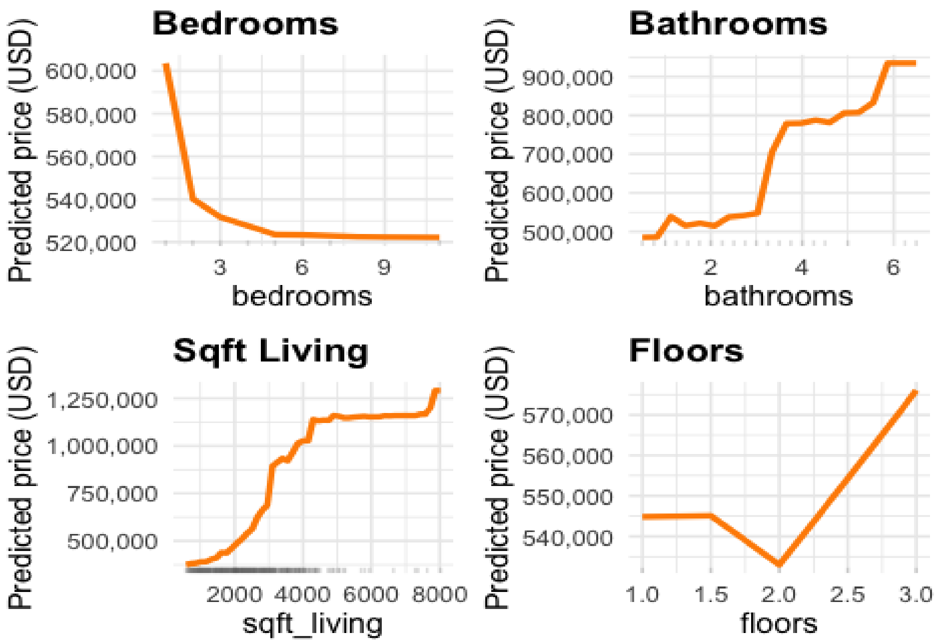
The 2D PDP heatmap reveals a clear joint pattern: the highest predicted bike rental values (lightest blue, approaching 5,000 rentals) occur when temperature is medium-to-high (normalized values between approximately 0.50 and 0.85) and humidity is moderate-to-low (below 0.60). Conversely, the darkest areas of the plot — representing the lowest predictions (around 3,000 rentals) — concentrate in two distinct regions: the leftmost columns (low temperature, any humidity) and the top rows (very high humidity, any temperature).

This result confirms and extends the findings from Exercise 1. The positive effect of temperature is robust across all humidity levels, but it is most pronounced when humidity is not extreme. Similarly, the negative effect of humidity is most damaging when temperatures are already low, producing the darkest cells in the upper-left corner of the heatmap. This interaction — where low temperature and high humidity jointly suppress demand far more than either factor alone — would not be detectable from the individual PDPs.

The scatter of real data points overlaid on the heatmap concentrates primarily in the central region of the plot, corresponding to normalized temperature values between 0.25 and 0.75 and humidity values between 0.40 and 0.80. This distribution indicates that the model's predictions are most reliable in this central region, where the training data provides adequate coverage. Predictions in the extreme corners of the heatmap — particularly very low temperature combined with very high humidity — are extrapolations based on fewer observations and should be interpreted with greater caution.

From an operational standpoint, the 2D PDP provides a practical decision-support tool: bike-sharing operators can use this visualization to anticipate demand under different combined weather conditions and plan fleet distribution or maintenance schedules accordingly.

5. Exercise 3 — PDP for House Price Prediction



Due to the large size of the dataset, a random subsample of 1,000 observations was used both for model training and for PDP computation. This sampling strategy is standard practice for large datasets and ensures tractable computation while preserving the overall distributional characteristics of the data. The model was trained with 500 trees. Partial Dependence Plots were computed for the four features explicitly requested in the exercise brief: bedrooms, bathrooms, sqft_living, and floors.

5.2 Results and Interpretation

Bedrooms

The PDP for the number of bedrooms produces a counterintuitive result: the predicted house price decreases as the number of bedrooms increases, dropping from approximately $610,000 for properties with one or two bedrooms to around $520,000 for properties with five or more bedrooms, where the curve stabilizes. This negative relationship challenges the naive assumption that more bedrooms always implies higher value.

A plausible economic interpretation is that the number of bedrooms alone is a poor proxy for property quality. In the King County market, a high bedroom count relative to the property size often signals that living space has been subdivided — a characteristic more common in budget housing than in luxury properties. Conversely, premium properties in desirable locations may feature fewer but larger, more architecturally significant rooms. The model has learned this pattern from the data. Furthermore, the rug plot on the x-axis confirms that the vast majority of observations fall between two and five bedrooms, meaning the model's predictions in this range are well-supported by data.

Bathrooms

In sharp contrast to bedrooms, the number of bathrooms displays a strongly positive relationship with predicted house prices. The PDP shows a gradual increase from approximately $490,000 for properties with one bathroom, accelerating sharply between three and four bathrooms, where the predicted price rises to nearly $800,000, before continuing upward toward $950,000 for properties with six or more bathrooms.

This pattern reflects well-established real estate market dynamics: the number of bathrooms is a reliable indicator of property size, quality, and the overall level of finish. Properties with multiple bathrooms cater to larger families or to buyers seeking luxury amenities, and are typically associated with higher-end construction standards. The non-linearity of the curve — with the steepest increase occurring between three and four bathrooms — suggests a market segmentation effect, where crossing the three-bathroom threshold signals a transition from standard to premium residential properties.

Sqft Living

Living area (sqft_living) is the most powerful predictor in this analysis, producing the largest range of predicted prices among the four variables studied. The PDP traces a near-monotonic positive relationship from approximately $350,000 for the smallest properties (around 500 sq ft) to over $1,250,000 for the largest (above 7,000 sq ft). The curve is not linear: it rises steeply between 1,000 and 4,000 square feet, then partially plateaus in the 4,000–6,000 sq ft range before resuming its upward trajectory.

This non-linearity is economically meaningful. At the lower end of the market, each additional square foot commands a high marginal premium as buyers compete for habitable space. In the mid-range, the market becomes more heterogeneous — properties of similar size may differ substantially in location, condition, and finishes — leading to a flatter price-size relationship. At the luxury end, very large properties again command strong premiums, but the rug plot reveals that very few observations exceed 6,000 sq ft, so predictions in this range carry greater uncertainty. Overall, sqft_living serves as the primary driver of predicted house prices in this model, consistent with decades of hedonic pricing research in real estate economics.

Floors

The number of floors exerts a more modest and non-monotonic influence on predicted prices. The PDP shows that single-story properties (1.0 floor) command predicted prices around $545,000, which remain essentially unchanged for 1.5-floor properties. At two floors, predicted prices dip slightly to approximately $535,000 before rising sharply for properties with 2.5 and 3 floors, reaching nearly $580,000.

The U-shaped pattern centered on two-floor properties likely reflects market composition effects. Two-story homes are the most common configuration in the dataset, and their prevalence means they span a wide range of price points — from modest family homes to upper-middle-class properties — which averages out to a moderate predicted price. Properties with 2.5 or 3 floors are considerably rarer and tend to represent either townhouses with specialized layouts or bespoke architectural designs, which command a premium. Despite this interesting non-linearity, the overall influence of floors on predicted price is substantially smaller than that of sqft_living or bathrooms.

5.3 Summary

The PDP analysis of the King County house price model identifies sqft_living and bathrooms as the two most influential predictors, both exhibiting strong positive relationships with price. The number of bedrooms surprisingly shows a negative effect, reflecting the nuanced way in which the real estate market values property configuration over mere room count. The number of floors has a moderate and non-linear effect, with multi-story specialty properties commanding a premium over the more common two-story format. These findings carry practical implications for property developers, appraisers, and investors seeking to understand the key value drivers in the King County residential market.

6. Conclusions

This report has demonstrated the practical value of Partial Dependence Plots as a model-agnostic interpretability tool applied to two distinct real-world prediction problems. By fitting Random Forest models — which are powerful but inherently opaque — and then using PDPs to illuminate their internal logic, it has been possible to extract clear, communicable insights about the relationships the models have learned.

In the bike rental domain, the analysis revealed that temperature is the dominant meteorological driver of demand, with an optimal range at moderate warmth and declining predictions at extremes. Humidity and wind speed both reduce demand, with humidity acting through a threshold mechanism and wind speed exerting a continuous negative effect. The temporal variable captures the system's growing adoption over time. The 2D PDP further revealed an interaction between temperature and humidity, showing that the combination of cold and humid conditions is particularly detrimental to demand — a finding that would have been missed with one-dimensional plots alone.

In the house pricing domain, living area emerged as the single most influential predictor, followed by the number of bathrooms. The counterintuitive negative effect of bedroom count highlights the importance of model interpretability: without PDPs, this unexpected relationship might have been overlooked or misattributed. The number of floors had a moderate non-linear effect, reflecting market segmentation between common two-story homes and more specialized multi-floor configurations.

From a methodological perspective, this exercise reinforces several important considerations for practitioners using PDPs: the importance of subsampling for computational tractability in large datasets, the need to inspect the distribution of real data points (via rug plots or scatter overlays) to assess prediction reliability across the feature range, and the value of extending to two-dimensional PDPs when interaction effects are suspected.

In conclusion, Partial Dependence Plots represent an essential component of any responsible machine learning workflow, enabling the kind of model transparency that is increasingly required by regulators, clients, and the broader public. The analyses presented here demonstrate that even complex ensemble models can be made interpretable and their learned relationships communicated clearly to non-technical stakeholders.